Week 9 Report

Name: HARINI M

Department: B.Sc. Data Science and Artificial Intelligence

Roll No: E25DA020

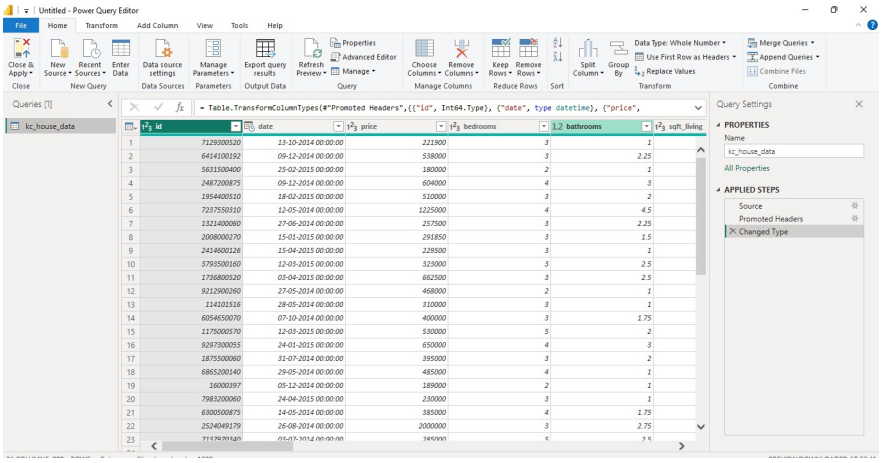
Week 9: Housing Data Cleaning & Transformation

Dataset: King County House Sales (21,613 rows, 21 columns) **Tool:** Power BI - Power Query Editor

Aim: Clean the housing sales data, extract date attributes, create derived columns, categorize properties by size and price, and load the data model.

1. Import data

Data was loaded into Power Query, headers were promoted and data types detected.

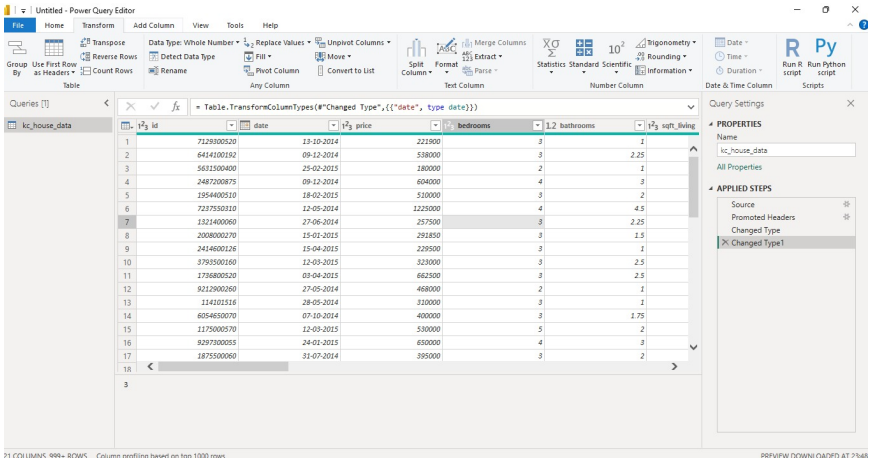


Data loaded in Power Query Editor

2. Parse sale date

The date column (Date/Time) was changed to Date type.

```
Table.TransformColumnTypes(#"Changed Type",{{"date", type date}})
```

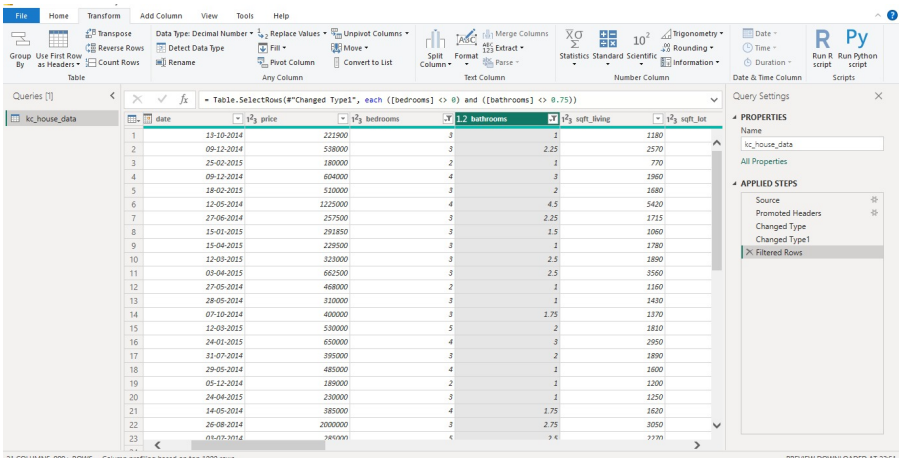


Date type changed

3. Filter invalid rows

Zero-bedroom and invalid bathroom rows were removed. Rows reduced from 21,613 to 21,529.

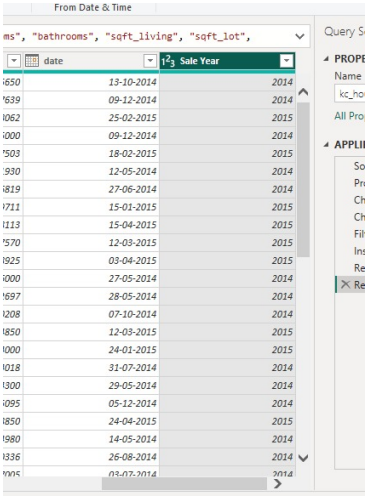
```
Table.SelectRows(#"Changed Type1", each ([bedrooms] <> 0) and ([bathrooms] <> 0.75))
```



Filtered Rows

4. Sale Year

Year was extracted from the date column (Add Column > Date > Year).



Sale Year column

5. Sale Month

Month number was extracted from the date column.

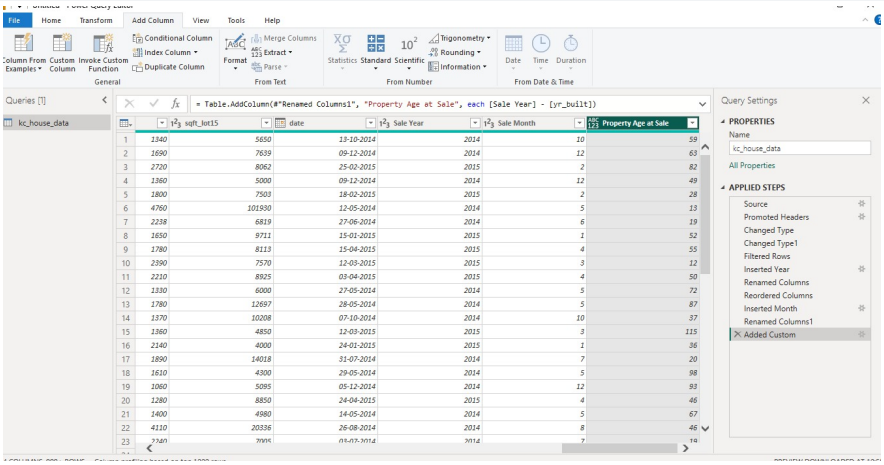
date	Sale Year	Sale Month
13-10-2014	2014	10
09-12-2014	2014	12
25-02-2015	2015	2
09-12-2014	2014	12
18-02-2015	2015	2
12-05-2014	2014	5
27-06-2014	2014	6
15-01-2015	2015	1
15-04-2015	2015	4
12-03-2015	2015	3
03-04-2015	2015	4
27-05-2014	2014	5
28-05-2014	2014	5
07-10-2014	2014	10
12-03-2015	2015	3
24-01-2015	2015	1
31-07-2014	2014	7
29-05-2014	2014	5
05-12-2014	2014	12
24-04-2015	2015	4
14-05-2014	2014	5
26-08-2014	2014	8
03-07-2014	2014	7

Sale Month column

6. Property Age at Sale

Age of the house at the time of sale.

```
Property Age at Sale = [Sale Year] - [yr_built]
```

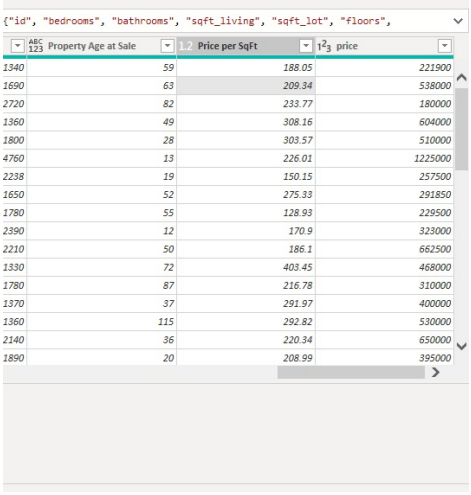


Property Age at Sale

7. Price per SqFt

Price divided by living area, rounded to 2 decimals.

```
Price per SqFt = Number.Round([price] / [sqft_living], 2)
```



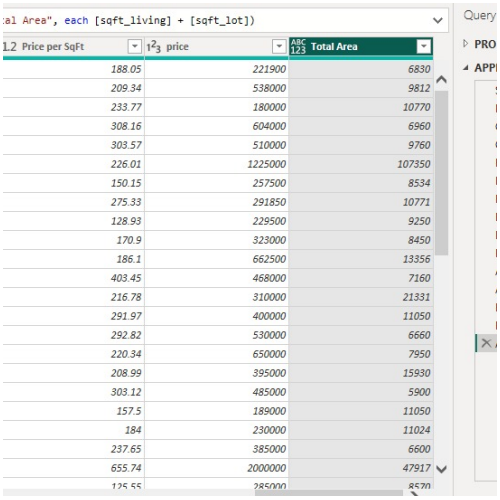
	Property Age at Sale	Price per SqFt	price
1340	59	188.05	221900
1690	63	209.34	538000
2720	82	233.77	180000
1360	49	308.16	604000
1800	28	303.57	510000
4760	13	226.01	1225000
2238	19	150.15	257500
1650	52	275.33	291850
1780	55	128.93	229500
2390	12	170.9	323000
2210	50	186.1	662500
1330	72	403.45	468000
1780	87	216.78	310000
1370	37	291.97	400000
1360	115	292.82	530000
2140	36	220.34	650000
1890	20	208.99	395000

Price per SqFt

8. Total Area

Living area plus lot area.

```
Total Area = [sqft_living] + [sqft_lot]
```



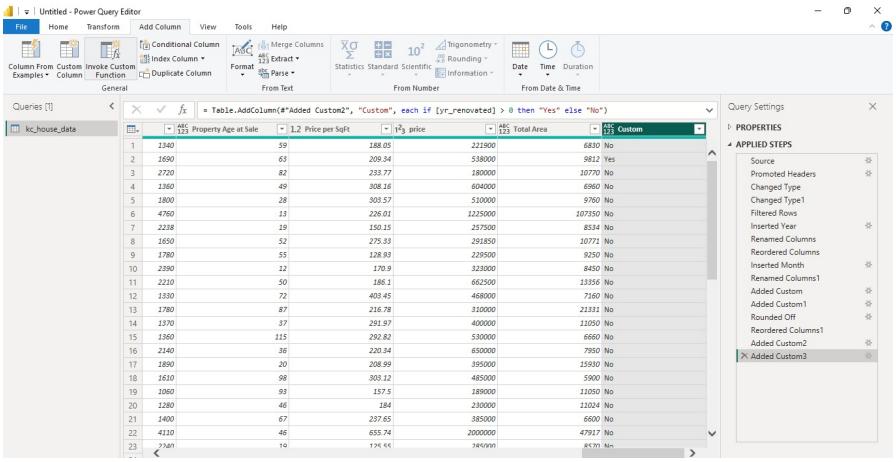
Price per SqFt	price	Total Area
188.05	221900	6830
209.34	538000	9812
233.77	180000	10770
308.16	604000	6960
303.57	510000	9760
226.01	1225000	107350
150.15	257500	8534
275.33	291850	10771
128.93	229500	9250
170.9	323000	8450
186.1	662500	13356
403.45	468000	7160
216.78	310000	21331
291.97	400000	11050
292.82	530000	6660
220.34	650000	7950
208.99	395000	15930
303.12	485000	5900
157.5	189000	11050
184	230000	11024
237.65	385000	6600
655.74	2000000	47917
125.55	285000	8570

Total Area

9. Is Renovated

Marks Yes if the house has a renovation year.

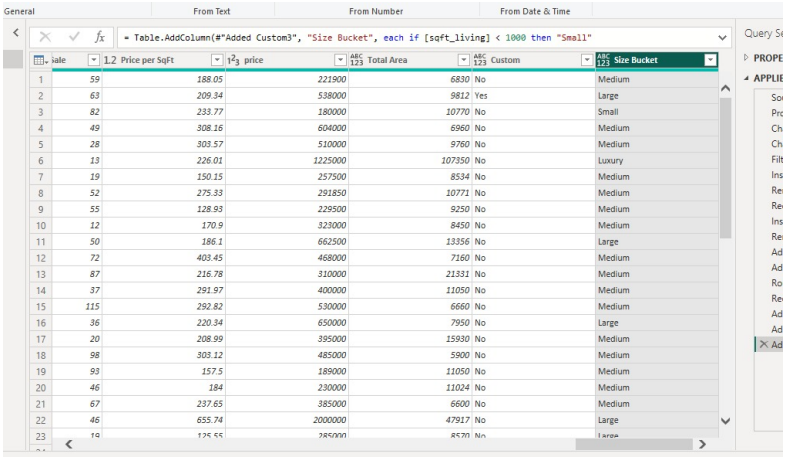
```
if [yr_renovated] > 0 then "Yes" else "No"
```



Is Renovated flag (column named Custom)

10. Size Bucket

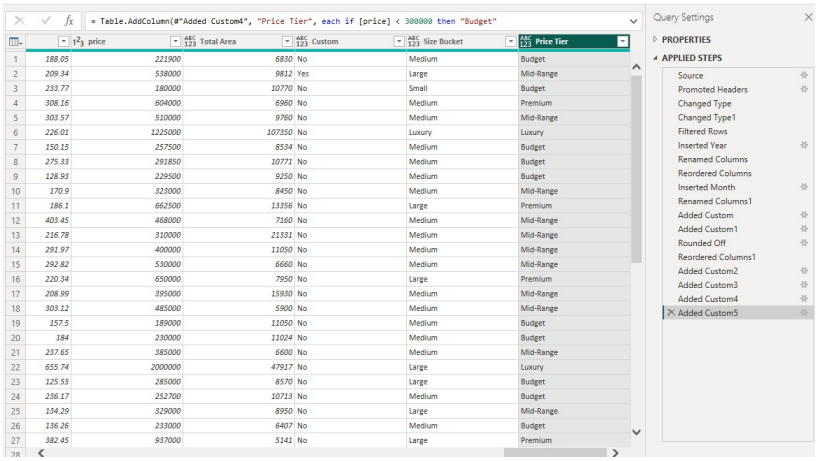
Small (<1000 sqft), Medium (1000-1999), Large (2000-3999), Luxury (4000+).



Size Bucket

11. Price Tier

Budget (<300k), Mid-Range (300k-599k), Premium (600k-999k), Luxury (1M+).

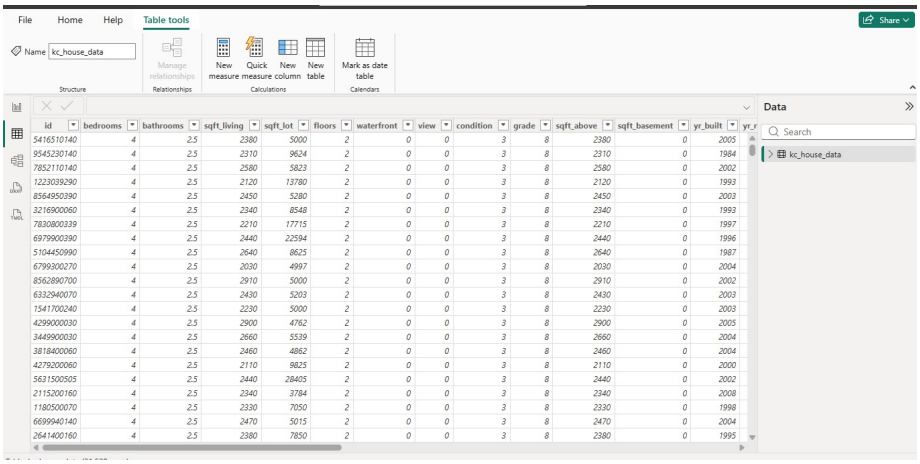


	price	Total Area	Custom	Size Bucket	Price Tier
1	188.05	221900	6830 No	Medium	Budget
2	209.34	538000	9812 Yes	Large	Mid-Range
3	233.77	180000	10770 No	Small	Budget
4	308.16	604000	6960 No	Medium	Premium
5	303.57	510000	9760 No	Medium	Mid-Range
6	226.01	1225000	107350 No	Luxury	Luxury
7	150.15	257500	8534 No	Medium	Budget
8	275.33	291850	10771 No	Medium	Budget
9	128.93	229500	9250 No	Medium	Budget
10	170.9	323000	8450 No	Medium	Mid-Range
11	186.1	662500	13356 No	Large	Premium
12	403.45	468000	7160 No	Medium	Mid-Range
13	216.78	310000	21331 No	Medium	Mid-Range
14	291.97	400000	11050 No	Medium	Mid-Range
15	292.82	530000	6660 No	Medium	Mid-Range
16	220.34	650000	7950 No	Large	Premium
17	208.99	395000	15930 No	Medium	Mid-Range
18	303.12	485000	5900 No	Medium	Mid-Range
19	157.5	189000	11050 No	Medium	Budget
20	184	230000	11024 No	Medium	Budget
21	237.65	385000	6600 No	Medium	Mid-Range
22	655.74	2000000	47917 No	Large	Luxury
23	125.55	285000	8570 No	Large	Budget
24	236.17	252700	10713 No	Medium	Budget
25	134.29	329000	8950 No	Large	Mid-Range
26	136.26	233000	6407 No	Medium	Budget
27	382.45	937000	5141 No	Large	Premium

Price Tier

12. Applied Steps and data model

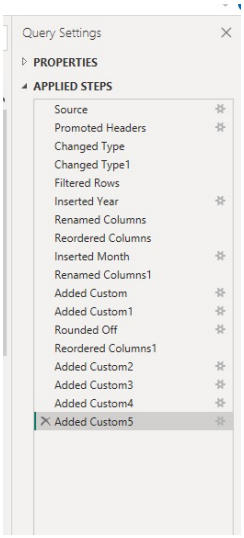
All steps are recorded in Applied Steps. After Close & Apply, the table (21,529 rows, 29 columns) was loaded into the data model.



id	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	condition	grade	sqft_above	sqft_basement	yr_built
5416510140	4	2.5	2380	5000	2	0	0	3	8	2380	0	2005
9545230140	4	2.5	2310	9624	2	0	0	3	8	2310	0	1984
7852110140	4	2.5	2580	5823	2	0	0	3	8	2580	0	2002
1223039290	4	2.5	2120	13780	2	0	0	3	8	2120	0	1993
8564950390	4	2.5	2450	5280	2	0	0	3	8	2450	0	2003
3216900060	4	2.5	2340	8548	2	0	0	3	8	2340	0	1993
7830800339	4	2.5	2210	17715	2	0	0	3	8	2210	0	1997
6979900390	4	2.5	2440	22594	2	0	0	3	8	2440	0	1996
5104450990	4	2.5	2640	8625	2	0	0	3	8	2640	0	1987
6799300270	4	2.5	2030	4997	2	0	0	3	8	2030	0	2004
8562890700	4	2.5	2910	5000	2	0	0	3	8	2910	0	2002
6332940070	4	2.5	2430	5203	2	0	0	3	8	2430	0	2003
1541700240	4	2.5	2230	5000	2	0	0	3	8	2230	0	2003
4299000030	4	2.5	2900	4762	2	0	0	3	8	2900	0	2005
3449900030	4	2.5	2660	5539	2	0	0	3	8	2660	0	2004
3818400060	4	2.5	2460	4862	2	0	0	3	8	2460	0	2004
4279200060	4	2.5	2110	9825	2	0	0	3	8	2110	0	2000
5631500505	4	2.5	2440	28405	2	0	0	3	8	2440	0	2002
2115200160	4	2.5	2340	3784	2	0	0	3	8	2340	0	2008
1180500070	4	2.5	2330	7050	2	0	0	3	8	2330	0	1998
6699940140	4	2.5	2470	5015	2	0	0	3	8	2470	0	2004
2641400160	4	2.5	2380	7850	2	0	0	3	8	2380	0	1995

Data model loaded

Applied Steps



Applied Steps list

Conclusion

The raw data of 21,613 records was cleaned to 21,529 records. Sale Year, Sale Month, Property Age at Sale, Price per SqFt, Total Area and Is Renovated were derived, and every property was grouped into a Size Bucket and Price Tier. The structured data model is ready for analysis in Power BI.